

7 Generalized Linear Models (GLM)

7.1 When the response is not normal

For the linear we assumed

$$y_i|x_i \sim N(\beta_0 + \beta_i x_i, \sigma)$$

However, $y_i|x_i$ may not follow a normal distribution. For this cases, we need a Generalized Linear Model (GLM).

The responds could also follow one of the following distributions:

- Normal (continuous)
- Poisson or negative binomial (count data)
- Gamma (positive continuous)
- Bernoulli (binary data)

7.1.1 Example for count data

Amphibian road kills

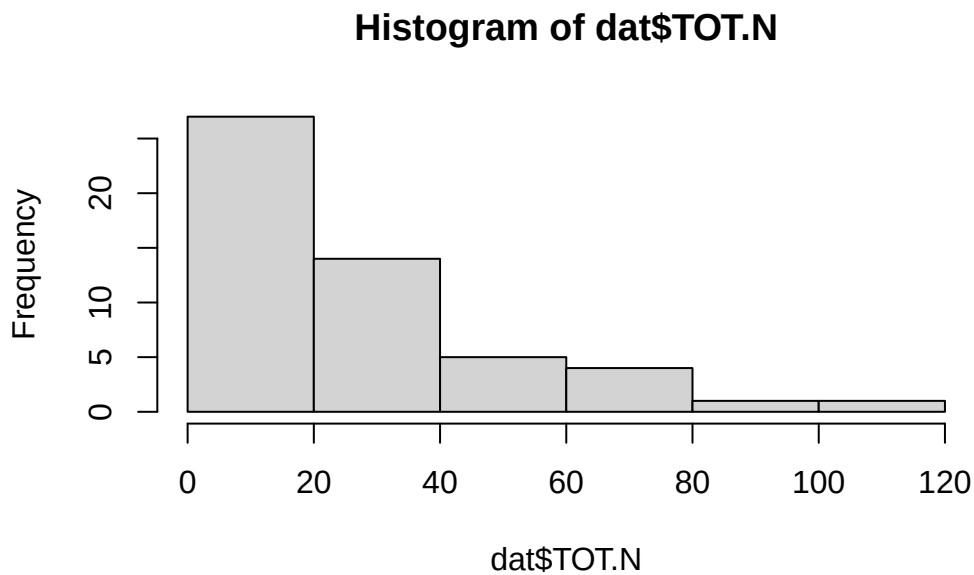
```
dat <- read.table(here::here("data/RoadKills.txt"), header = TRUE)
head(dat)
```

	Sector	X	Y	BufoCalamita	TOT.N	S.RICH	OPEN.L	OLIVE	MONT.S	MONT
1	1	260181	256546	5	22	3	22.684	60.333	0.000	0.653
2	2	259914	256124	1	14	4	24.657	40.832	0.000	0.161
3	3	259672	255688	40	65	6	30.121	23.710	0.258	10.918
4	4	259454	255238	27	55	5	50.277	14.940	1.783	26.454
5	5	259307	254763	67	88	4	43.609	35.353	2.431	11.330
6	6	259189	254277	56	104	7	31.385	17.666	0.000	43.678
	POLIC	SHRUB	URBAN	WAT.RES	L.WAT.C	L.D.ROAD	L.P.ROAD	D.WAT.RES	D.WAT.COUR	
1	4.811	0.406	7.787	0.043	0.583	3330.189	1.975	252.113	735.000	
2	2.224	0.735	27.150	0.182	1.419	2587.498	1.761	139.573	134.052	

3	1.946	0.474	28.086	0.453	2.005	2149.651	1.250	59.168	269.029
4	0.625	0.607	0.831	0.026	1.924	4222.983	0.666	277.842	48.751
5	0.791	0.173	2.452	0.000	2.167	2219.302	0.653	967.808	126.102
6	0.054	0.325	2.730	0.039	2.391	1005.629	1.309	560.000	344.444

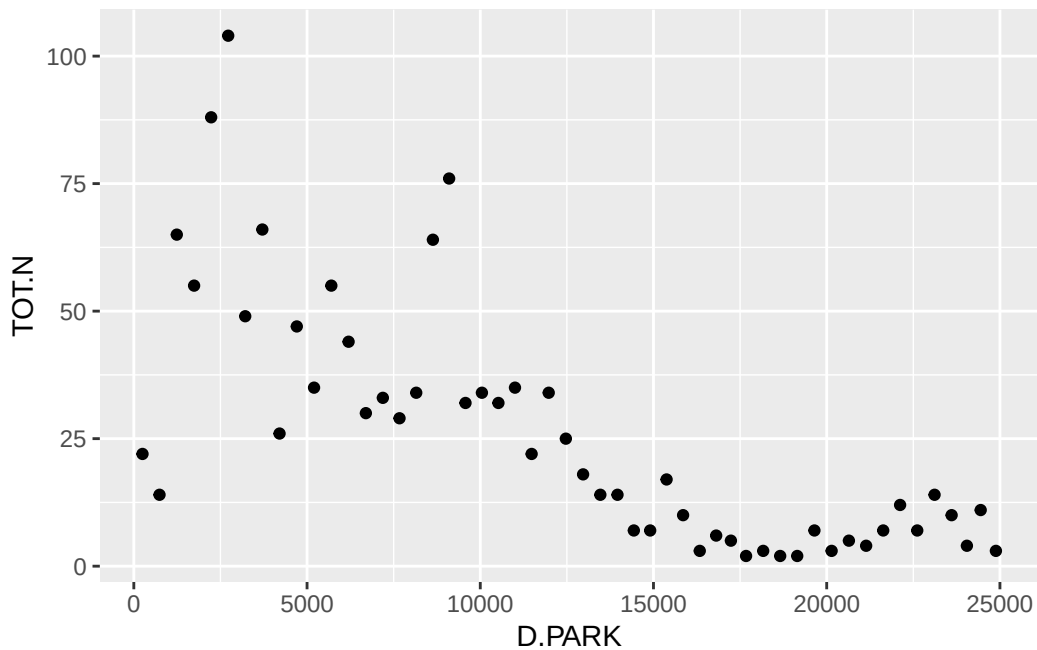
	D.PARK	N.PATCH	P.EDGE	L.SDI
1	250.214	122	553.936	1.801
2	741.179	96	457.142	1.886
3	1240.080	67	432.360	1.930
4	1739.885	63	421.292	1.865
5	2232.130	59	407.573	1.818
6	2724.089	49	420.289	1.799

```
hist(dat$TOT.N)
```



We want to model the response (TOT.N) as a Poisson distribution with rate λ that changes according to the distance to a park.

```
ggplot(dat, aes(D.PARK, TOT.N)) + geom_point()
```



7.1.2 Example for presence absence data

Tuberculosis in wild boar, this can be seen as a Bernoulli random variable (a boar is either infected or not).

```
dat <- read.table(here::here("data/Boar.txt"), header = TRUE)
table(dat$Tb)
```

```
0    1
314 285
```

```
ggplot(dat, aes(LengthCT, Tb)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm") +
  geom_smooth(method = "glm",
              method.args = list(family = "binomial"), col = "red")
```

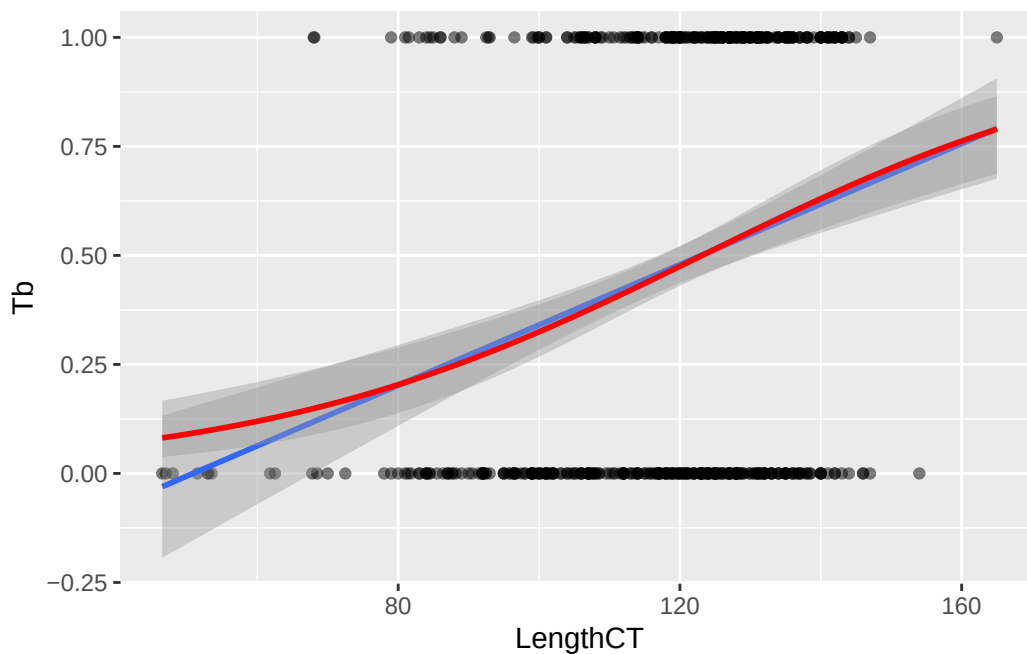
`geom\_smooth()` using formula = 'y ~ x'

Warning: Removed 149 rows containing non-finite outside the scale range (``stat_smooth()``).

``geom_smooth()`` using formula = 'y ~ x'

Warning: Removed 149 rows containing non-finite outside the scale range (``stat_smooth()``).

Warning: Removed 149 rows containing missing values or values outside the scale range (``geom_point()``).



7.2 Structure of GLMs

A GLM consists of three components:

1. The distribution of the response.
2. Specification of the systematic part of the model (i.e., the linear predictor), usually denoted as η .
3. A link $g(\cdot)$ between the expectation of the response distribution and the systematic part of the model. This is also called the **link function**.

Some common link functions are:

Table 15.1 Some Common Link Functions and Their Inverses

<i>Link</i>	$\eta_i = g(\mu_i)$	$\mu_i = g^{-1}(\eta_i)$
Identity	μ_i	η_i
Log	$\log_e \mu_i$	e^{η_i}
Inverse	μ_i^{-1}	η_i^{-1}
Inverse-square	μ_i^{-2}	$\eta_i^{-1/2}$
Square-root	$\sqrt{\mu_i}$	η_i^2
Logit	$\log_e \frac{\mu_i}{1 - \mu_i}$	$\frac{1}{1 + e^{-\eta_i}}$
Probit	$\Phi^{-1}(\mu_i)$	$\Phi(\eta_i)$
Log-log	$-\log_e[-\log_e(\mu_i)]$	$\exp[-\exp(-\eta_i)]$
Complementary log-log	$\log_e[-\log_e(1 - \mu_i)]$	$1 - \exp[-\exp(\eta_i)]$

NOTE: μ_i is the expected value of the response; η_i is the linear predictor; and $\Phi(\cdot)$ is the cumulative distribution function of the standard-normal distribution.

Figure 7.1: From Fox 2016 Applied Regression Analysis and Generalized Linear Models; page 419.

Table 15.2 Canonical Link, Response Range, and Conditional Variance Function for Exponential Families

<i>Family</i>	<i>Canonical Link</i>	<i>Range of Y_i</i>	$V(Y_i \eta_i)$
Gaussian	Identity	$(-\infty, +\infty)$	ϕ
Binomial	Logit	$0, 1, \dots, n_i$	$\mu_i(1 - \mu_i)$
Poisson	Log	$0, 1, 2, \dots$	μ_i
Gamma	Inverse	$(0, \infty)$	$\phi \mu_i^2$
Inverse-Gaussian	Inverse-square	$(0, \infty)$	$\phi \mu_i^3$

NOTE: ϕ is the dispersion parameter, η_i is the linear predictor, and μ_i is the expectation of Y_i (the response). In the binomial family, n_i is the number of trials.

Figure 7.2: From Fox 2016 Applied Regression Analysis and Generalized Linear Models; page 421.

Note, the linear model is a special case of a GLM with the identity function as the link function.

7.3 Example Poisson GLM

Count data occurs whenever the response is a count of some quantities. For example number of accidents, number of individuals, number of years, etc.

A possible model for such data is the Poisson distribution, with the following probability mass function:

$$f(y; \lambda) = \frac{\lambda^y \exp(-\lambda)}{y!}$$

With:

- $y \geq 0$ and y being an integer, and
- λ being the rate and $\lambda \geq 0$.

Note, that for a Poisson distribution: $E(y) = Var(y) = \lambda$.

1. The distribution of the response: **Poisson distribution**
2. Specification of the systematic part of the model $\eta_i = \beta_0 + \beta_1 x_i$
3. Link function: log

That means:

- $\eta_i = \beta_0 + \beta_1 x_i$
- $\log(\lambda_i) = \eta_i$ or $\lambda_i = \exp(\eta_i)$.
- Finally, $y_i \sim P(\lambda_i)$

In R there is a function called `glm`.

```
dat <- read.table(here::here("data/RoadKills.txt"), header = TRUE)
pr1 <- glm(TOT.N ~ D.PARK, data = dat, family = poisson(link = "log"))
summary(pr1)
```

Call:

```
glm(formula = TOT.N ~ D.PARK, family = poisson(link = "log"),
    data = dat)
```

Coefficients:

```

              Estimate Std. Error z value Pr(>|z|)
(Intercept)  4.316e+00  4.322e-02   99.87  <2e-16 ***
D.PARK       -1.059e-04  4.387e-06  -24.13  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

(Dispersion parameter for poisson family taken to be 1)

```

Null deviance: 1071.4  on 51  degrees of freedom
Residual deviance:  390.9  on 50  degrees of freedom
AIC: 634.29

```

Number of Fisher Scoring iterations: 4

How many road kills would we expect 1000 m away from a park?

```
coef(pr1) %*% rbind(1, 1000)
```

```

      [,1]
[1,] 4.210634

```

This is on the linear scale, we have to apply $g(\cdot)^{-1}$ to get back at the response scale. Since we used the log link function, the inverse can be obtained with `exp`.

```
exp(coef(pr1) %*% rbind(1, 1000))
```

```

      [,1]
[1,] 67.39928

```

Note, that the `predict()` function also has an argument `type` that can take the value `'link'` or `'response'`.

By default type is set to `"link"`:

```
predict(pr1, data.frame(D.PARK = 1000))
```

```

      1
4.210634

```

```
predict(pr1, data.frame(D.PARK = 1000), type = "response")
```

```
1  
67.39928
```

7.3.1 Interpretation

1. Transform coefficients
2. Using figures

```
pr0 <- glm(TOT.N ~ D.PARK, data = dat, family = poisson())
```

7.3.1.1 Transform coefficients

We used the model

$$\begin{aligned}\log \mu &= \beta_0 + \beta_1 x_1 \\ \mu &= \exp(\beta_0 + \beta_1 x_1) \\ &= \exp(\beta_0) \times \exp(\beta_1 x_1) \\ &= \exp(\beta_0) \times \exp(\beta_1)^{x_1}\end{aligned}$$

This means that the effect of β_1 is multiplicative. Increasing x_1 by one unit, changes y by the factor $\exp(\beta_1)$.

```
p1 <- predict(pr0, data.frame(D.PARK = 1001), type = "response")  
p2 <- predict(pr0, data.frame(D.PARK = 1002), type = "response")  
p2 / p1
```

```
1  
0.9998942
```

This is the same as

```
exp(coef(pr0)[2])
```

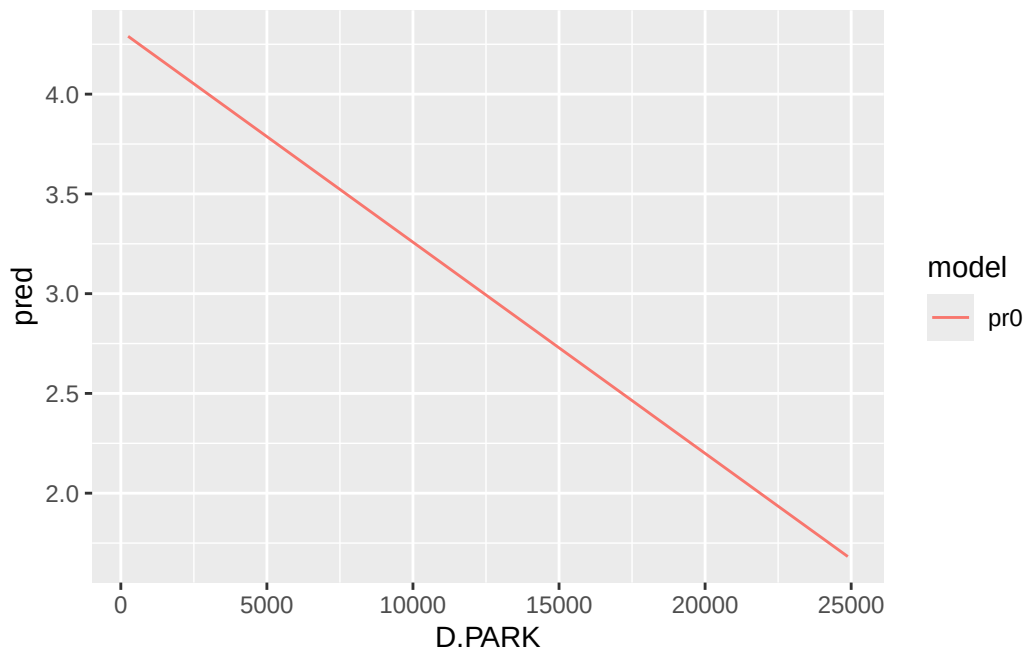
```
D.PARK  
0.9998942
```


7.3.1.2 Using figures

Note, I use here the package `modelr` for making predictions. This is between `ggeffects` and the base R `predict()` function.

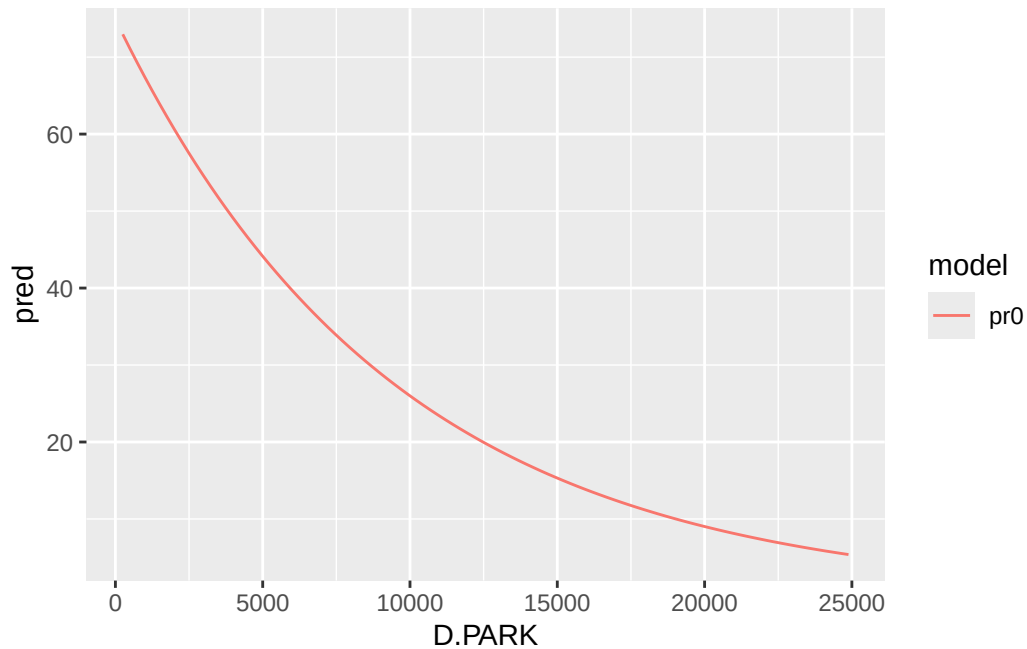
I use first the function `data_grid()` as a more convenient alternative to `expand.grid()`. Then `gather_predictions()` is slightly more convenient than `predict()`.

```
library(modelr)
data_grid(dat,
  D.PARK = seq_range(D.PARK, 100),
  N.PATCH = typical(N.PATCH),
  S.RICH = typical(S.RICH)) %>%
gather_predictions(pr0) %>%
ggplot(aes(D.PARK, pred, col = model)) + geom_line()
```



We can also specify the response type.

```
data_grid(dat,
  D.PARK = seq_range(D.PARK, 100),
  N.PATCH = typical(N.PATCH),
  S.RICH = typical(S.RICH)) %>%
gather_predictions(pr0, type = "response") %>%
ggplot(aes(D.PARK, pred, col = model)) + geom_line()
```



7.4 The logistic regression

The logistic regression can be use, if we have a response that can take on two values: 0 or 1, yes or no, male or female, ...

What we model, is the expectation π of a binomial distribution.

1. The distribution of the response: **Binomial distribution**
2. Specification of the systematic part of the model $\eta_i = \beta_0 + \beta_1 x_i$
3. Link function: *logit*

Thus, we have

$$\text{logit}(\pi_i) = \eta_i$$

this means that

$$\pi_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

7.4.1 Interpretation of parameters

For the logistic regression, the coefficients can be interpreted in terms of odds ratio.

```
dat <- read.table(here::here("data/Boar.txt"), header = TRUE) %>%
  mutate(sex = factor(SEX, labels = c("male", "female")))
m1 <- glm(Tb ~ sex + LengthCT, data = dat, family = binomial())
summary(m1)
```

Call:

```
glm(formula = Tb ~ sex + LengthCT, family = binomial(), data = dat)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.02584	0.70309	-5.726	1.03e-08 ***
sexfemale	-0.20098	0.19101	-1.052	0.293
LengthCT	0.03359	0.00578	5.811	6.21e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 681.25 on 493 degrees of freedom
Residual deviance: 640.12 on 491 degrees of freedom
(163 observations deleted due to missingness)
AIC: 646.12

Number of Fisher Scoring iterations: 4

The odds are defined as

$$\frac{\pi_i}{1 - \pi_i}$$

Using the inverse logit $1/(1 + \exp(-\eta))$ we get

$$\frac{1/(1 + \exp(-\beta_0 + \beta_1 x))}{1/(1 + \exp(-\beta_0 + \beta_1(x + 1)))} = \exp(\beta_1)$$

- If $\beta_1 = 0$ the odds do not change.
- If $\beta_1 < 0$ the odds decrease.
- If $\beta_1 > 0$ the odds increase.

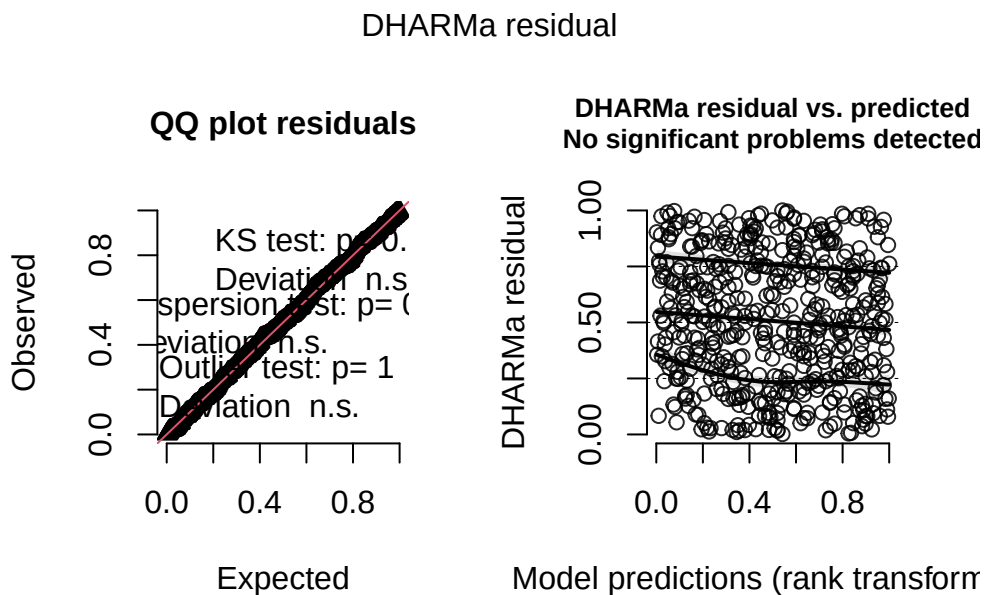
7.4.2 How well does the model fit the data?

Residual diagnostics is not straight-forward any more. One options to check residuals are simulations from the model (these are called randomized quantile residuals). The package DHARMa implements this.

```
library(DHARMa)
```

This is DHARMa 0.4.7. For overview type '?DHARMa'. For recent changes, type `news(package = 'DHARMa')`.

```
plot(simulateResiduals(m1))
```



🔥 Exercise 1: Fitting a GLM

Linden and Rohloff (2015) studied white-headed woodpeckers in California (USA) using point counts. The file `woodp_occ.csv` contains data from three visits at 66 sites. For each site and visit we know the if a woodpecker was present or not (columns `y.1`, `y.2`, `y.3`), the day of the year (columns `date.1`, `date.2`, `date.3`) and the snag density (the number of tree snags per ha; column `snags`). Tree snags are standing but dead or dying trees. Only use the first session of data collection for this exercise (i.e., `y.1`).

1. What would be a suitable model for this kind of data (i.e., think of the distribution of y)?
2. Fit the model that you decided on in 1).
3. Give an interpretation of the model results.

4. Create a suitable plot to communicate the results.
5. Validate your model.
6. *Optional*: Can you fit the same model with Stan?